📘 Session 4 – Data Manipulation Using Functions

Session Objectives

By the end of this session, students will be able to:

1. Apply Excel functions to **rank, sort, filter, and transpose** data.
2. Use **conversion functions** to manipulate data types and formats.
3. Apply functions like **TEXT()**, **VALUE()**, **DOLLAR()**, **FIXED()**, and **DATEVALUE()** in real-life scenarios.

🔹 1. Overview of Data Manipulation

Data manipulation means **transforming or restructuring data** to make it easier to analyze and interpret.

In Excel, you can manipulate data using **formulas, functions, and references** instead of manually editing data.

Example Use Case:
In the `Employee_Data` sheet, you might want to:

- Rank employees by performance,
- Filter only top performers,
- Or convert salary values into text for reporting.

🔹 2. Functions to Rank, Sort, Filter, and Transpose Data

🧮 RANK() Function

Used to assign a ranking position to numbers within a dataset.

Syntax:

```
=RANK(number, ref, [order])
```

- `number` : the cell to rank
- `ref` : the range of numbers
- `order` : 0 (descending) or 1 (ascending)

Example:
In `Employee_Data` , rank employees by performance:

```
=RANK(F2, $F$2:$F$21, 0)
```

This ranks the highest performance score as **1**.

🔄 SORT() Function (Excel 365 and later)

Sorts data dynamically without changing the original dataset.

Syntax:

```
=SORT(array, [sort_index], [sort_order], [by_col])
```

Example:
Sort employees by salary (highest to lowest):

```
=SORT(A2:F21, 4, -1)
```

🔍 FILTER() Function

Extracts data that meets specific criteria.

Syntax:

```
=FILTER(array, include, [if_empty])
```

Example:
Show only employees from the **Sales** department:

```
=FILTER(A2:F21, C2:C21="Sales")
```

🔁 TRANSPOSE() Function

Converts rows to columns and columns to rows.

Syntax:

```
=TRANSPOSE(array)
```

Example:
Transpose product names from row to column in `Product_Data`:

```
=TRANSPOSE(B2:B16)
```

🧠 Demonstration (Hands-on Practice)

Using `Employee_Data`:

1. Rank employees by performance (create a new column “Rank”).
2. Sort the table dynamically by salary.
3. Filter the list to show only HR department employees.
4. Transpose department names to appear horizontally.

💡 Real-life Applications

Function	Application Example
RANK()	Ranking exam scores or KPIs
SORT()	Arranging products by sales
FILTER()	Displaying active customers only
TRANSPOSE()	Converting vertical reports into horizontal summaries

🔷 3. Conversion Functions

Conversion functions change **data types or formats**—for example, turning numbers into formatted text or converting text dates into real date values.

🧩 TEXT() Function

Converts a numeric value to text in a specific format.

Syntax:

```
=TEXT(value, format_text)
```

Example:
Format salary as currency:

```
=TEXT(D2, "$#,##0.00")
```

Result: `$45,000.00`

🔢 VALUE() Function

Converts a text string representing a number into a numeric value.

Syntax:

```
=VALUE(text)
```

Example:
Convert “52000” (text) to a real number:

```
=VALUE("52000")
```

Useful when importing data from external sources where numbers come in as text.

💲 DOLLAR() Function

Converts a number to text using currency formatting with a dollar sign (or localized equivalent).

Syntax:

```
=DOLLAR(number, [decimals])
```

Example:

```
=DOLLAR(45000, 2)
```

Result: `$45,000.00`

📏 FIXED() Function

Rounds a number to a specified number of decimals and converts it to text.

Syntax:

```
=FIXED(number, [decimals], [no_commas])
```

Example:

```
=FIXED(12345.678, 2, FALSE)
```

Result: `12,345.68`

📅 DATEVALUE() Function

Converts a date stored as text into a valid Excel date serial number.

Syntax:

```
=DATEVALUE(date_text)
```

Example:

```
=DATEVALUE("2024-05-20")
```

Result: `45435` (Excel serial for the date).
You can then format it back to date format.

🧠 Demonstration (Hands-on)

In `Employee_Data`:

1. Create a new column **Salary_Text** using:

   ```
   =TEXT(D2, "₦#,##0.00")
   ```

2. Convert any text salary value back to number using:

   ```
   =VALUE(E2)
   ```

3. Display salary in fixed decimal places:

   ```
   =FIXED(D2, 2)
   ```

🔹 4. Class Exercises

🧩 Basic Level

1. In `Sales_Data`, rank all sales by **Total_Sales**.
2. Convert all *Unit_Price* values to currency using the **DOLLAR()** function.

⚙️ Intermediate Level

1. In `Employee_Data`, filter only employees in the **IT** department with a salary above ₦60,000.
2. Use **TEXT()** to format the salary column as "₦#,##0.00".

🚀 Advanced Level

1. In `Product_Data`, use **FILTER()** to show products with *Stock* < *Reorder_Level*.
2. Use **RANK()** and **SORT()** together to display the top 5 high-priced products.
3. Apply **DATEVALUE()** to convert import dates from text to date values for reporting.

🔹 5. Summary

Concept	Description
RANK(), SORT(), FILTER(), TRANSPOSE()	Used for rearranging, extracting, and restructuring data
TEXT(), VALUE(), DOLLAR(), FIXED(), DATEVALUE()	Used for data conversion and formatting
Data Manipulation	Makes datasets more readable and analysis-ready

🧾 Assignment

Use your workbook to:

1. Create a **Top 5 Performers** table using **RANK() + FILTER()**.
2. Format salaries using **TEXT()** and display as currency.
3. Export your result as a new Excel sheet titled **"Manipulated_Data"**.